

Technical audit of an algorithmic decision-support system:

1. Background

Our chosen Algorithmic Decision-Support System (ADS) is the public Kaggle notebook implementation "Kfold LightGBM," developed for the Home Credit Default Risk competition. The ADS is a credit-risk prediction system that uses application-level information, demographic variables, financial characteristics, and historical credit behavior to estimate whether a loan applicant is likely to experience repayment difficulty. The unit of analysis is one loan application, and the model outputs a predicted risk score for TARGET = 1.

The main purpose of the ADS is predictive. It is designed to distinguish applicants who are more likely to experience repayment difficulty from those who are less likely to do so. Although the Kaggle notebook is not a production lending system, its output can still function as decision support. A lender could use the score to rank applicants by risk, identify cases for closer review, or inform approval and denial decisions.

This creates a tradeoff between risk control and credit access. A stricter model may better identify applicants who are likely to struggle with repayment, but it may also incorrectly flag creditworthy applicants as high risk. A more permissive model may expand access to credit, but it may miss genuinely risky cases. This is why the ADS is worth auditing: strong overall prediction does not guarantee fair or harmless outcomes across groups.

2. Input and output

The ADS uses the Home Credit Default Risk dataset. The main applicant-level files are application_train.csv and application_test.csv. The notebook also uses auxiliary credit-history tables: bureau.csv, bureau_balance.csv, previous_application.csv, POS_CASH_balance.csv, installments_payments.csv, and credit_card_balance.csv. These tables are aggregated to the applicant level using SK_ID_CURR and merged into the final modeling table.

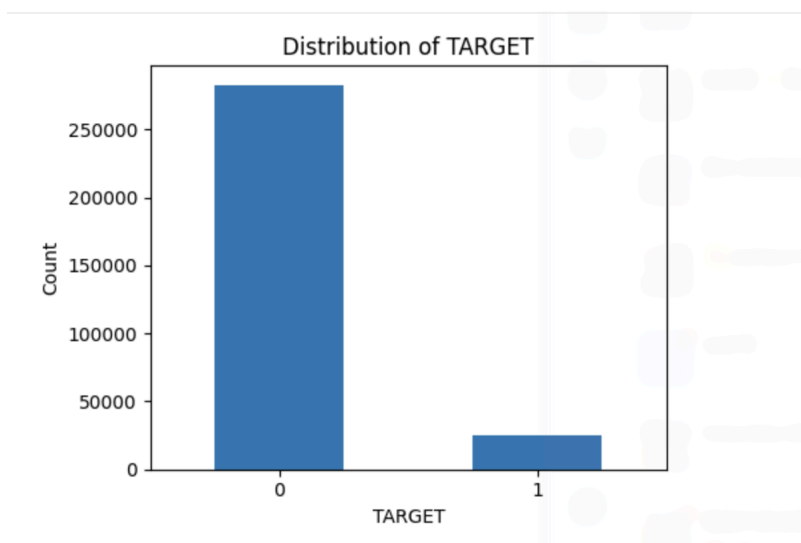
Table 1. Raw data sources used by the audited ADS

File	Rows	Columns	Role in the ADS
application_train.csv	307,511	122	Main training table with applicant features and target label

application_test.csv	48,744	121	Test table with applicant features but no target label
bureau.csv	1,716,428	17	External credit bureau records
bureau_balance.csv	27,299,925	3	Monthly bureau balance history
previous_application.csv	1,670,214	37	Historical previous loan applications
POS_CASH_balance.csv	10,001,358	8	POS/cash loan balance records
installments_payments.csv	13,605,401	8	Installment payment history
credit_card_balance.csv	3,840,312	23	Credit card balance history

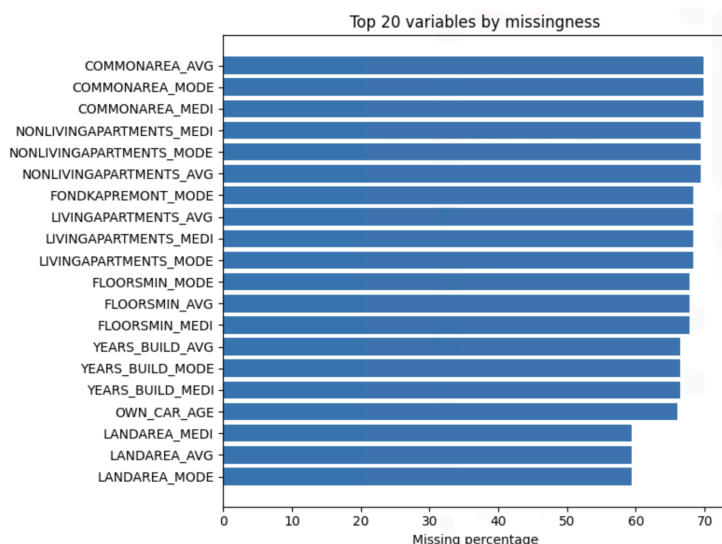
The final unit of analysis is the applicant-level loan application. The raw training application table has shape (307511, 122), while the raw test application table has shape (48744, 121). After preprocessing, feature engineering, and aggregation, the final training matrix has shape (307507, 720) and the final test matrix has shape (48744, 720).

The main training table contains mixed data types: 65 float columns, 41 integer columns, and 16 object columns. The target variable is highly imbalanced. In application_train.csv, TARGET = 0 appears in 282,686 cases, or 91.927%, while TARGET = 1 appears in 24,825 cases, or 8.073%. This imbalance matters because accuracy alone can make a model look stronger than it really is.



Missingness is also substantial, especially in housing and property variables. COMMONAREA_AVG, COMMONAREA_MODE, and COMMONAREA_MEDI each have about 69.87% missing values. NONLIVINGAPARTMENTS_* variables have about 69.43% missingness, and LIVINGAPARTMENTS_* variables have about 68.35% missingness. The

ADS therefore operates with incomplete applicant information rather than a fully observed feature set.



Among selected continuous variables, the strongest correlations appear among loan-size features. `AMT_CREDIT` and `AMT_GOODS_PRICE` are highly correlated at 0.987. `AMT_CREDIT` and `AMT_ANNUITY` have correlation 0.770, and `AMT_ANNUITY` and `AMT_GOODS_PRICE` have correlation 0.775. The `EXT_SOURCE_*` variables are only moderately correlated with one another and weakly correlated with the main monetary variables, suggesting that the model combines loan-size information with distinct external score information.

The ADS output is a probability-like risk score estimating whether an applicant belongs to the positive class, `TARGET = 1`. In practical terms, it estimates repayment-difficulty risk. The notebook writes test-set predictions to `lightgbm.csv`, but conceptually the output is a continuous score that can rank applicants or support threshold-based lending decisions.

3. Implementation and validation

The audited notebook implements a full applicant-level tabular machine-learning pipeline. It reads the main application files, cleans and encodes the data, engineers additional features, aggregates auxiliary relational tables to the applicant level, and trains a LightGBM binary classifier.

Preprocessing begins by combining `application_train.csv` and `application_test.csv` so both receive the same transformations. The notebook removes records where `CODE_GENDER = 'XNA'` and replaces the sentinel value 365243 in `DAYS_EMPLOYED` with missing values. It factorizes selected binary categorical variables such as `CODE_GENDER`, `FLAG_OWN_CAR`, and `FLAG_OWN_REALTY`, while broader categorical variables are one-hot encoded.

The notebook also creates applicant-level engineered features, including credit-to-annuity, credit-to-goods, and credit-to-income ratios, along with summary features based on `EXT_SOURCE_*` variables. It then processes `bureau.csv`, `bureau_balance.csv`, `previous_application.csv`, `POS_CASH_balance.csv`, `installments_payments.csv`, and `credit_card_balance.csv`. For each auxiliary table, it computes grouped statistics such as maxima, means, sums, counts, and variances by `SK_ID_CURR`, then merges those features into the main applicant-level table. This allows the model to use both current application information and summarized historical credit behavior.

The model is LightGBM, a gradient-boosted decision-tree method used for binary classification. Validation uses 7-fold KFold cross-validation with ROC-AUC as the main metric. In the full run, the fold-level AUC values were 0.792309, 0.792948, 0.793479, 0.790008, 0.793425, 0.789649, and 0.787180. The overall AUC was 0.791222.

These results show that the model meets its basic predictive goal at the aggregate level. It ranks applicants by repayment-difficulty risk better than chance. Still, aggregate AUC does not show whether the model behaves similarly across groups, whether false positive and false negative rates are uneven, or whether the model depends on features that may act as proxies for demographic or socioeconomic status. Those questions are addressed in the audit outcomes.

4. Outcomes:

4.1 Overall model performance

We first evaluated aggregate predictive performance, then compared performance across subpopulations. The model achieved an overall ROC-AUC of 0.791222 using 7-fold KFold cross-validation.

ROC-AUC is appropriate because the target is highly imbalanced: only about 8.1% of applicants have `TARGET = 1`. Accuracy alone would be misleading because a model could look accurate by favoring the majority class. ROC-AUC evaluates how well the model ranks higher-risk applicants above lower-risk applicants, which is more relevant for credit-risk scoring.

Aggregate ROC-AUC is not enough for auditing this ADS. Since the score could influence lending decisions such as approval, denial, or closer review, uneven errors across groups could affect access to credit differently. For that reason, we audited subgroup performance across gender, age group, and education level.

4.2 Threshold-based decision framing

The LightGBM model outputs a risk score, not a direct approval or denial. To audit it as a decision-support system, we defined a threshold rule: applicants in the top 10% of predicted risk scores are flagged as high risk or closer review. This does not mean the model automatically denies credit. It creates a decision-relevant category that lets us examine how a lender might use the score in practice.

Under this rule, the risk threshold was 0.191965, and the overall high-risk flag rate was 0.10. We measured subgroup performance using ROC-AUC, accuracy, high-risk flag rate, precision, true positive rate (TPR), false positive rate (FPR), and false negative rate (FNR). A false positive means an applicant who did not experience repayment difficulty was still flagged as high risk, which could push a creditworthy applicant toward unnecessary scrutiny or denial. A false negative means an applicant who did experience repayment difficulty was not flagged, which creates lender risk and may lead to harmful lending if applicants are offered credit they cannot repay.

Because the model score could affect access to credit or determine which applicants receive closer review, we interpret subgroup differences as potential allocation disparities, not just technical performance gaps. A group with a higher FPR may face more unnecessary scrutiny, while a group with a higher FNR may be exposed to lending decisions that underestimate repayment difficulty.

4.3 Accuracy across subpopulations

Gender

Female applicants had a slightly higher AUC than male applicants, but the threshold outcomes show the more important difference.

Group	N	AUC	Accuracy	High-risk flag rate	Precision	TPR	FPR	FNR
Female	202,448	0.7901	0.8990	0.0775	0.3001	0.3324	0.0584	0.6676
Male	105,059	0.7827	0.8451	0.1433	0.3134	0.4428	0.1095	0.5572

Male applicants were flagged at 14.33%, compared to 7.75% for female applicants. Male applicants also had a higher FPR, meaning non-defaulting male applicants were more likely to be

incorrectly flagged as high risk. The model is more aggressive in flagging male applicants, which improves recall but increases false positives.

Age group:

Age showed stronger disparities than gender.

Group	N	AUC	Accuracy	High-risk flag rate	Precision	TPR	FPR	FNR
Under 30	45,184	0.7653	0.8142	0.1790	0.3007	0.4703	0.1414	0.5297
30-44	123,736	0.7930	0.8655	0.1185	0.3125	0.4113	0.0895	0.5887
45-59	103,286	0.7881	0.9081	0.0675	0.3052	0.3141	0.0502	0.6859
60+	35,301	0.7547	0.9379	0.0290	0.2780	0.1642	0.0220	0.8358

Applicants under 30 were flagged at 17.90%, while applicants 60 and older were flagged at only 2.90%. The under-30 group also had the highest FPR, meaning younger applicants without repayment difficulty were most likely to be incorrectly flagged. The 60+ group had the highest FNR at 0.8358, meaning the model missed many true repayment-difficulty cases among older applicants. From an audit perspective, the model appears to over-flag younger applicants and under-identify risk among older applicants.

Education level:

Group	N	AUC	Accuracy	High-risk flag rate	Precision	TPR	FPR	FNR
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Secondary / secondary special	218,389	0.7865	0.8663	0.1173	0.3110	0.4081	0.0887	0.5919
Higher education	74,862	0.7900	0.9262	0.0457	0.2789	0.2380	0.0348	0.7620
Other / lower / incomplete	14,256	0.7783	0.8603	0.1202	0.2958	0.3924	0.0931	0.6076

Applicants with higher education were flagged at 4.57%, compared to 11.73% for secondary education and 12.02% for other, lower, or incomplete education. The higher education group had the lowest FPR but the highest FNR, meaning the model missed more true repayment-difficulty cases in that group. The model appears to treat higher education as a strong signal of lower risk. That reduces false positive flags for some applicants but may under-detect actual repayment difficulty within that group.

4.4 Fairness analysis

We used three fairness-relevant comparisons: high-risk flag rate, FPR, and TPR. High-risk flag rate relates to demographic parity because it shows how often each group receives the high-risk label. FPR matters because false positives can harm creditworthy applicants through added scrutiny or denial. TPR relates to whether the model identifies true repayment-difficulty cases at similar rates across groups.

The results show that the ADS does not behave evenly across groups. Male applicants are flagged more often and have a higher FPR. Applicants under 30 have the highest flag rate and highest FPR, while applicants 60 and older have the highest FNR. Higher-education applicants are flagged less often but have a higher FNR.

These patterns do not prove intentional discrimination. They do show that the same threshold creates different types of harm across groups. For younger applicants and applicants with secondary or lower/incomplete education, the main concern is over-flagging and unnecessary scrutiny. For older applicants and higher-education applicants, the concern is under-identifying true repayment difficulty.

4.5 Additional audit method: interpretability through feature importance

We used feature importance as an additional audit method because it helps show what types of variables drive the model's predictions and whether those variables may act as proxies for age, socioeconomic status, or prior access to credit.

The most important features in the full model included NEW_CREDIT_TO_ANNUITY_RATIO, NEW_EXT_SOURCES_MEAN, DAYS_BIRTH, EXT_SOURCE_3, EXT_SOURCE_2, EXT_SOURCE_1, DAYS_ID_PUBLISH, AMT_ANNUITY, DAYS_EMPLOYED, and ACTIVE_DAYS_CREDIT_MEAN.

The model relies heavily on credit-payment ratios, external source scores, age-related variables, employment timing, annuity amount, and historical credit activity. This is expected for a credit-risk model, but some features may also reflect socioeconomic status, age, financial stability, or unequal access to formal credit systems. DAYS_BIRTH directly captures age, while external source scores and credit-history variables may encode past inequality in access to credit.

This supports the subgroup findings. The model is not only using neutral loan-size features. It is also relying on applicant characteristics and historical credit signals that can vary across demographic and socioeconomic groups.

4.6 Overall outcome assessment

The ADS meets its basic predictive goal. A cross-validated ROC-AUC of 0.791222 means the model ranks applicants by repayment-difficulty risk meaningfully better than chance, and the implementation is technically functional.

That is not enough for deployment. The threshold analysis shows differences across gender, age, and education. Some groups are flagged at much higher rates, while others have elevated false negative rates. In a real lending setting, these differences could become unequal access to credit, unequal scrutiny, or harmful lending decisions.

We would not treat this ADS as ready for direct deployment based on AUC alone. A lender would need to monitor subgroup performance, justify the decision threshold, examine proxy variables, and add fairness-aware review procedures before using the model for consequential credit decisions.

5. Summary

The Home Credit dataset is appropriate for auditing this ADS because it directly relates to a credit decision setting. The model predicts repayment difficulty, and that prediction could influence closer review, loan terms, or denial. The data also contains applicant-level variables such as gender, age, education, family status, housing type, income, employment history, and prior credit behavior, which makes subgroup auditing possible. At the same time, the data has

limits. It comes from a specific lending context, includes high missingness in some feature groups, and contains proxy variables that may reflect existing social and economic inequality.

The implementation is technically strong but only partly fair. The LightGBM model achieved an overall ROC-AUC of 0.791222, which shows reasonable ranking performance. ROC-AUC is the right primary accuracy metric because the target is imbalanced. For fairness auditing, we used high-risk flag rate, FPR, FNR, and TPR because they connect model behavior to stakeholder harms. Lenders may care about AUC and recall because they want to detect repayment risk. Applicants may care more about false positives because those can lead to unnecessary scrutiny or reduced credit access. Regulators and auditors may care about differences in error rates across groups because those differences can become allocation disparities.

The fairness results show uneven behavior. Male applicants and applicants under 30 were flagged more often and had higher false positive rates. Applicants 60 and older had the highest false negative rate. Applicants with higher education were flagged less often but had a higher false negative rate, while applicants with secondary or lower/incomplete education were flagged more often. These results do not prove intentional discrimination, but they show that the model's risk labels and errors are not evenly distributed.

We would not be comfortable deploying this ADS as an automatic credit approval or denial system. It may be useful as decision support for ranking or closer review, but final decisions should involve human review, threshold justification, subgroup monitoring, and an appeals process. Credit decisions affect financial access, housing stability, business opportunities, and long-term mobility, so a model that looks accurate overall can still create unfair burdens if its errors fall unevenly across groups.

Before responsible use, the lender should conduct regular subgroup audits, test multiple thresholds, review proxy variables, and document how the model affects applicants. High-risk predictions should trigger further review, not automatic rejection. Without these safeguards, the ADS may be useful as a predictive tool, but not fair or transparent enough for consequential lending decisions.

6. Project contributions

Both partners contributed to the project and reviewed the final report, notebook, slides, and presentation materials.

Kevin primarily worked on the technical foundation of the project. He identified and reproduced the Home Credit Default Risk LightGBM notebook, downloaded and organized the Kaggle data, reviewed the raw input files, described the input and output structure, explained the

preprocessing and feature engineering steps, and documented the model implementation and validation results.

Bilal primarily worked on the audit and fairness analysis portion of the project. He reviewed the project requirements and professor feedback, reframed the ADS around credit decision consequences, added the subgroup audit analysis, generated the gender, age, and education subgroup metric tables, interpreted the fairness and error-rate results, and wrote the Outcomes and Summary sections. He also helped polish the final report structure and connect the analysis to the ethical concerns of credit-risk decision support.

References:

Home Credit Default Risk. Kaggle competition dataset

<https://www.kaggle.com/competitions/home-credit-default-risk/data>

Ashish Patel. "Kfold LightGBM." Kaggle notebook.

<https://www.kaggle.com/code/ashishpatel26/kfold-lightgbm/notebook>